

Seokjun Choi

[sək-dʒu:n tʃwe] (approx. “Suk-jun Choi”)

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Korean (Native) · English (Advanced)

Research Overview

Research Interests: Computational Appearance, Inverse Rendering, Polarization Imaging, Appearance Fabrication, and Visuo-Tactile Sensing

Technical Skills: Differentiable rendering, deep learning, optical system design, and calibrated image acquisition

My research explores how computational imaging and graphics can bridge **digital representations and the physical world**. I have developed practical **display-camera inverse rendering systems** for measuring and reconstructing real-world geometry and reflectance, with first-author publications at **CVPR 2024, ICCV 2025, and ACM TOG (SIGGRAPH Asia 2026)**. My current work extends this direction from **observing physical appearance** toward **fabricating material appearance** and **sensing physical interaction**, through polarimetric appearance acquisition, BRDF fabrication, and visuo-tactile sensing.

Education

POSTECH (Pohang University of Science and Technology), Pohang, Korea Feb. 2022 – Feb. 2028 (Expected)

Ph.D. Student, Department of Computer Science and Engineering

Advisor: [Seung-Hwan Baek](#)

Member of [Computational Imaging Group](#), in [Computer Graphics Lab](#)

Chung-Ang University, Seoul, Korea

Feb. 2015 – Aug. 2021

B.S. in Integrative Engineering and B.S. in Software

GPA: 4.11/4.5, Magna Cum Laude

Publications

Seokjun Choi*, Yunseong Moon*, Kaizhang Kang, Hoon-Gyu Chung, Jin-Nyeong Kim, Giljoo Nam, and Seung-Hwan Baek.

“Snapshot Polarimetric Display Inverse Rendering.”

ACM Transactions on Graphics, 45(6), Article 201, 2026. [**Proc. SIGGRAPH Asia 2026**]

doi:10.1145/3842531 *Equal contribution.

Seokjun Choi, Hoon-Gyu Chung, Yujin Jeon, Giljoo Nam, and Seung-Hwan Baek.

“A Real-world Display Inverse Rendering Dataset.”

Proceedings of the IEEE/CVF International Conference on Computer Vision [ICCV 2025]

Hoon-Gyu Chung, **Seokjun Choi**, and Seung-Hwan Baek.

“Differentiable Inverse Rendering with Interpretable Basis BRDFs.”

Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition [CVPR 2025]

Juhyung Choi, Jinnyeong Kim, **Seokjun Choi**, Jinwoo Lee, Samuel Brucker, Mario Bijelic, Felix Heide, and Seung-Hwan Baek.

“Dual Exposure Stereo for Extended Dynamic Range 3D Imaging.”

Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition [CVPR 2025]

Seokjun Choi, Seungwoo Yoon, Giljoo Nam, Seungyong Lee, and Seung-Hwan Baek.
“Differentiable Display Photometric Stereo.”
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition [CVPR 2024]

Hoon-Gyu Chung, **Seokjun Choi**, and Seung-Hwan Baek.
“Differentiable Point-based Inverse Rendering.”
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition [CVPR 2024]

Suhyun Shin, **Seokjun Choi**, Felix Heide, and Seung-Hwan Baek.
“Dispersed Structured Light for Hyperspectral 3D Imaging.”
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition [CVPR 2024]

Reviewer Services

Conference Reviewer

- SIGGRAPH Asia 2026
- IEEE VR 2026
- NeurIPS 2025, 2026

Patents

U.S. Patent Application

- Seung-Hwan Baek, **Seokjun Choi**, Seungwoo Yoon, Seungyong Lee, Jinwoo Park, Nahyup Kang, and Jiyeon Kim.
“Method of determining illumination pattern for three-dimensional scene and method and apparatus for modeling three-dimensional scene.”
US-2024-0394964-A1.
- Seung-Hwan Baek, Suhyun Shin, **Seokjun Choi**
“System, method, and apparatus for 3D hyperstpectral imaging.”
US-2025-0369800-A1.

Honors and Awards

First Prize, Outstanding Paper Award	2025
BK21 End-to-End Big Data Innovation Talent Development Program. POSTECH, Pohang, Korea.	
Top-ranked admission scholarship and full tuition support	2015–2016, 2018–2020
Awarded for all enrolled semesters. Chung-Ang University, Seoul, Korea.	

Teaching

Teaching Assistant @ POSTECH

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|---|------------------|
| • POSCO AI Expert | 2022, 2023, 2024 |
| • CSED490F : Computational Signal Processing | Fall 2025 |
| • CSED233 : Data Structure | Fall 2023 |
| • CSED700G : Computational Imaging | Spring 2022 |

Research Projects

Neural Rendering for Acquisition of Realistic 3D Assets

Sponsored by SAIT (Samsung Advanced Institute of Technology).

Mar. 2023 – Dec. 2023

Performance Analysis of 2D Image-Based Semantic Segmentation Algorithm

Sponsored by ETRI (Electronics and Telecommunications Research Institute).

Jun. 2022 – Nov. 2022

Visual Memory Network-based Cognitive Imitation

Developed network compression algorithms for surveillance cameras.

Sponsored by ETRI (Electronics and Telecommunications Research Institute).

Apr. 2020 – Nov. 2020

Experiences

Republic of Korea Air Force

Sergeant in charge of administration, 256th Tactical Airlift Squadron, 15th Combat Group.
15th Special Activity Wing, Republic of Korea Air Force.

Jul. 2016 – Jul. 2018